Results: Language, Topics, and Embeddings

RQ3: Topical and Linguistic Structure I

Text analysis operates over titles, abstracts, and (when available) full text. A TF-IDF representation over a 500-feature vocabulary feeds non-negative matrix factorization, which extracts 5 latent topics cross-cutting the subfield taxonomy. The top vocabulary terms are: modafinil, treatment, study, results, patients, effects, sleep, used, clinical, use, drug, studies, mg, using, sleepiness, disorder, associated, cognitive, narcolepsy, significant.

Table 3. NMF topics extracted from the corpus.

Topic	Top terms
0	modafinil, mg, kg, effects, rats, mice, induced, sleep
1	fatigue, patients, placebo, modafinil, scale, armodafinil, treatment, depression
2	h4, methods, results, conclusion, 95, analysis, ci, risk
3	use, cognitive, drugs, adhd, studies, drug, cocaine, methylphenidate
4	sleep, narcolepsy, sleepiness, cataplexy, daytime, patients, eds, excessive

RQ3: Topical and Linguistic Structure II

The topics reveal the thematic structure of the literature: Topic 0 centres on cognitive enhancement and neuroenhancement; Topic 1 addresses ADHD treatment and clinical evidence; Topic 2 covers pharmacological dose-response studies (including animal models); Topic 3 focuses on sleep disorders (narcolepsy, excessive daytime sleepiness); and Topic 4 addresses fatigue in psychiatric populations. These topics cross-cut the keyword-based subfield taxonomy, revealing connections that the explicit classification does not capture.

RQ3: Topical and Linguistic Structure III

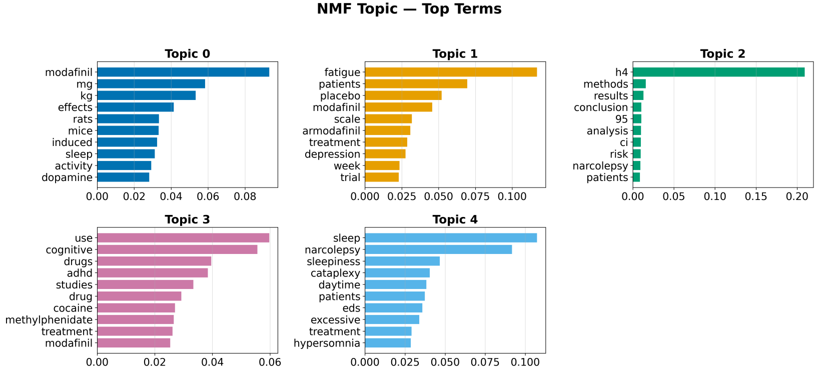
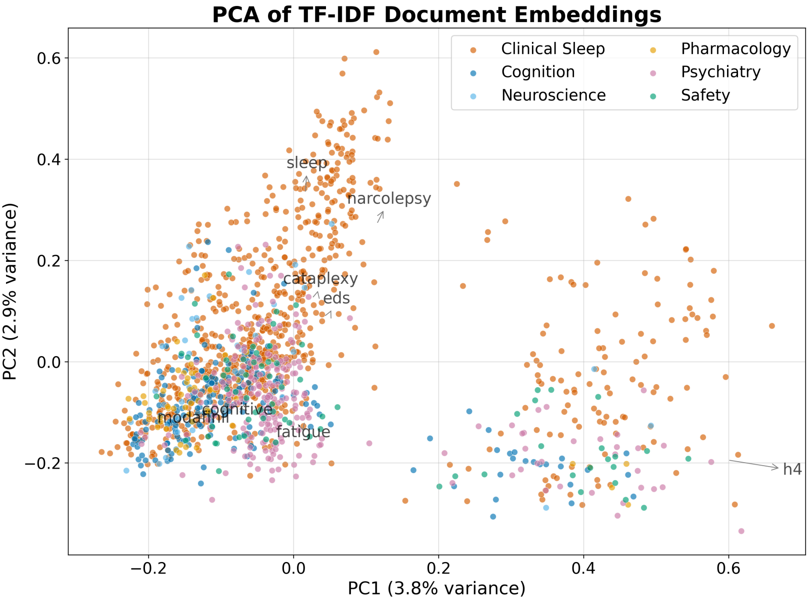


Figure 1: Topic-term bar charts for Modafinil. Each panel shows the top weighted terms for one of 5 NMF topics, with bar length proportional to the topic-term weight in the $\mathbf{H}$ matrix.

Document Embeddings I

Offline deterministic embeddings (TF-IDF followed by truncated SVD) place every document in a shared 50-dimensional vector space. Embedding the same text twice yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

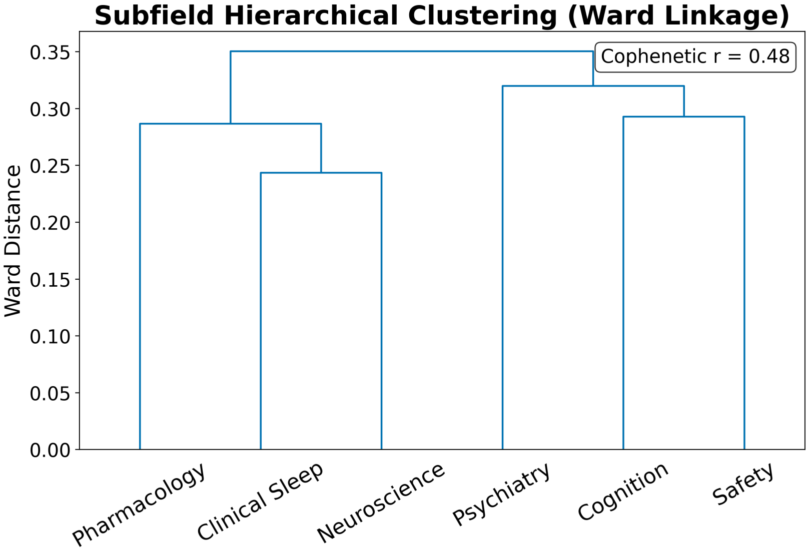
Document Embeddings II



Document Embeddings III

Figure 2: PCA projection of document embeddings for Modafinil. Each point represents one document projected onto the first two principal components of the TF-IDF/SVD embedding. Colours indicate subfield assignment, showing how the topical geography relates to the keyword taxonomy.

Document Embeddings IV

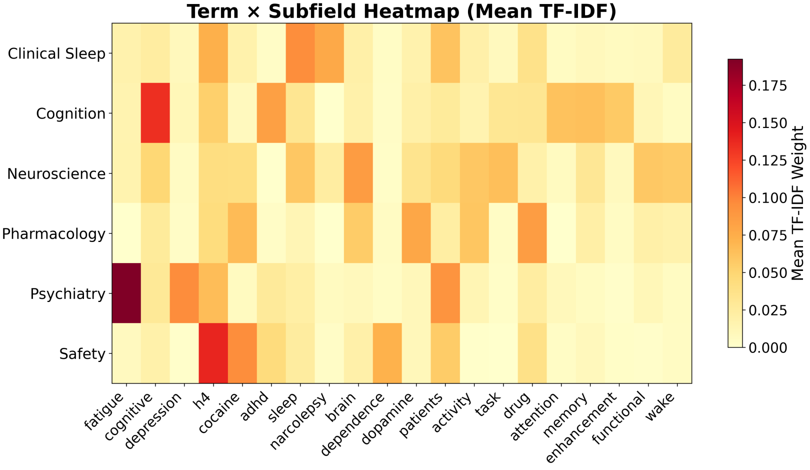


Document Embeddings V

Figure 3: Hierarchical clustering dendrogram of document embeddings. The tree shows the similarity structure of the corpus: documents that join low in the tree are semantically similar, while high-level splits separate the major topical clusters.

Term Analysis I

The TF-IDF term heatmap reveals which terms discriminate between subfields: terms with high between-subfield variance (rather than high global mean) are selected for display.



Term Analysis II

Figure 4: Term heatmap for Modafinil. Each cell shows the mean TF-IDF weight of a term within a subfield. Terms are selected by between-subfield variance to highlight discriminative vocabulary rather than globally frequent terms.

Named Entity Analysis I

Named entity extraction over the 1437 abstracts identified 30 unique entities. The most frequent entities reflect the clinical and pharmacological vocabulary of the modafinil literature.

Table 4. Top named entities in abstracts.

Entity	Frequency
ADHD	392
CI	343
EDS	326
OSA	272
MOD	181
ESS	148
RESULTS	148
MS	145
DAT	134
NT1	131
SD	130
CONCLUSIONS	129
MD	102
CE	101
IH	101

Named Entity Analysis II

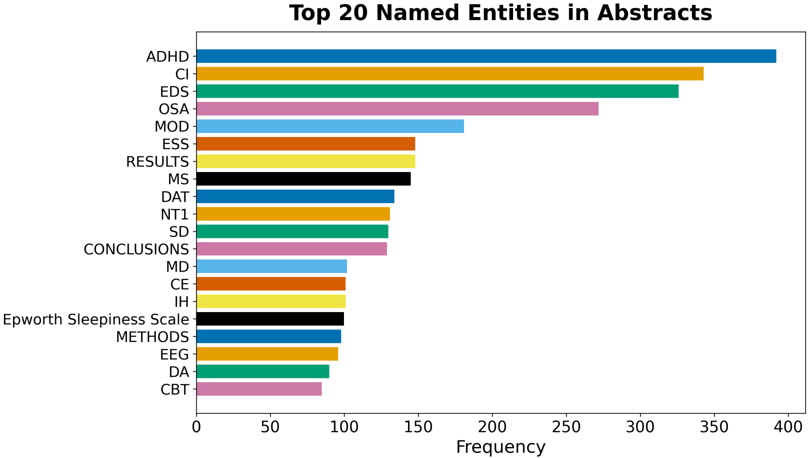


Figure 5: Top named entities for Modafinil. The horizontal bar chart shows the 20 most frequently extracted named entities from abstracts, revealing the dominant drugs, conditions, and concepts in the literature.

Named Entity Analysis III

Table 5. Top keyphrases by TF-IDF score.

Keyphrase	Score
available	0.3333
abstract	0.3333
abstract available	0.3333
content	0.1053
access	0.1053
sub	0.0881
md	0.0870
jama	0.0826
cleveland	0.0763
modafinil	0.0741
depression	0.0741
cognitive	0.0714
substance	0.0694
drug	0.0667
conditions	0.0667

Embedding Similarity and Clustering I

The TF-IDF/SVD embeddings place every document in a 50-dimensional vector space. K-means clustering with $k = 5$ clusters partitions the corpus into topically coherent groups. The top similar document pairs, ranked by cosine similarity, reveal the most closely related works in the corpus.

Table 6. Top 10 most similar document pairs.

Paper A	Paper B	Similarity
doi:10.1176/appi.ajp.163.12.21	doi:10.1176/ajp.2006.163.12.21	1.0000
doi:10.1176/ajp.2006.163.12.21	doi:10.1176/appi.ajp.163.12.21	1.0000
doi:10.1345/aph.1h302	doi:10.1136/bcr.08.2011.4652	0.9660
doi:10.4088/jcp.09m05900gry	doi:10.1186/s40345-015-0034-0	0.9566
doi:10.1111/bdi.12859	doi:10.1111/acps.12712	0.9558
doi:10.1111/j.1360-0443.2008.0	doi:10.1111/j.1465-3362.2012.0	0.9536
doi:10.1513/annalsats.202006-6	doi:10.3760/cma.j.cn112147-202	0.9532
doi:10.1164/ajrccm.164.9.21030	doi:10.1093/sleep/28.4.464	0.9513
doi:10.1093/sleep/28.4.464	doi:10.1164/ajrccm.164.9.21030	0.9513
doi:10.1002/cncr.25083	doi:10.1200/jco.2005.23.16_sup	0.9507

Embedding Similarity and Clustering II

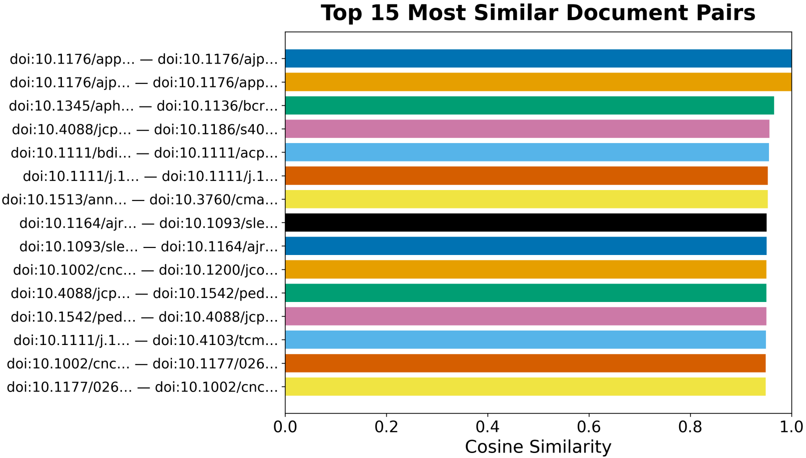
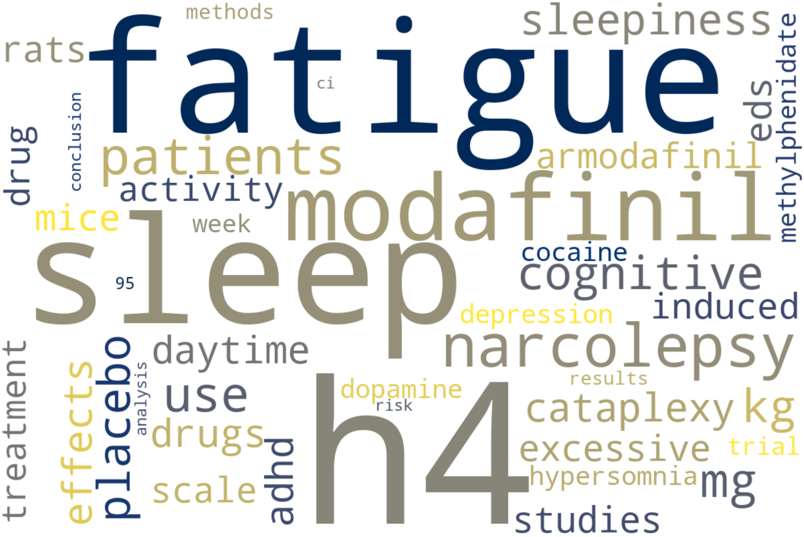


Figure 6: Document similarity for Modafinil. The horizontal bar chart shows the 15 most similar document pairs ranked by cosine similarity of their TF-IDF/SVD embeddings. High-similarity pairs share topical and lexical content.

Embedding Similarity and Clustering III

Literature Corpus — Term Cloud



Embedding Similarity and Clustering IV

Figure 7: Term cloud for Modafinil. Term sizes are proportional to their TF-IDF weights across the corpus, providing a visual summary of the dominant vocabulary.

Embedding Similarity and Clustering V